PRODUCT USER MANUAL

For WAVE_GLO_PHY_SPC-FWK_L4_NRT_014_004

Issue: 1.0

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RECORD TABLE

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1.0	2025/05/15	All	New format of the PUM including the information of the last version of the CMEMS-WAV-PUM-014-001-002-003-004-009-V3. EIS November 2025 : Modification of Product name and dataset name	Marine De Carlo	Patricia Zunino

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GLOSSARY AND ABBREVIATIONS

ADD	Architecture and Design Document
AL	Saral/AltiKa
C2	Cryosat-2
CalVal	Calibration/Validation
CF	Climate Forecast (convention for NetCDF)
CFO	CFOSAT
CLS	Collecte Localisation Satellite
CMEMS	Copernicus Marine Environment Monitoring Service
DT	Delayed Time
DU	Dissemination Unit
GDR	Geophysical Data Record
GOP	Geophysical Ocean Product
HY-2B	Hai Yang 2B
HY-2C	Hai Yang 2C
IGDR	Intermediate Geophysical Data Record
J3	Jason-3
KPI	Key Production Indicator
L2/L2P/L3/L4	Level 2 / Level 2 Plus / Level 3 / Level 4
LRM	Low Resolution Mode
NetCDF	Network Common Data Form
NOP	NRT Ocean Product
NRT	Near Real Time
OE	Orbit Error
OGDR	Operational Geophysical data record
OLA	Operational Level Agreement
PIT	Product Information Table
PLRM	Pseudo LRM
PQ	Product Quality
PU	Production Unit
PUM	Product User Manual
QUID	QUality Information Document
RMS	Root Mean Square
RPC	Remote Protocol Client
RT	Real Time
S1A	Sentinel-1A
S1B	Sentinel-1B
S3A	Sentinel-3A
S3B	Sentinel-3B
S6A	Sentinel-6A
SAR	Synthetic Aperture Radar
SARIn	SAR Interferometric
SPC	SPeCtral
SRD	System Requirement Document
SWH	Significant Wave Height
SWON	SWOT nadir
TAC	Thematic Assembly Center
TDS	Thredds Data Server

VAVH	Visual AVerage Height
WMS	Web Map Service

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is one of the user manuals for the Copernicus Marine Service Near-Real-Time satellite wave products produced by the WAVE-TAC (Thematic Assembly Centre).

The WAVE TAC (Thematic Assembly Centre) is one of the eight TAC of the Copernicus Marine Service. The data produced in the frame of this TAC are generated by different processing systems including data from altimeter and other wave measuring devices (SAR, SWIM) for all the Copernicus missions (Sentinel-6A, Sentinel-3A/B, Sentinel-1A/B) and other collaborative or opportunity missions (e.g. : Jason-3, HaiYang-2 missions, Saral/AltiKa, Cryosat-2, CFOSAT).

This Product User Manual (PUM) details the following Level-4 product which is delivered in -Real-Time by the WAVE-TAC

- WAVE_GLO_WAV_L4_SPC-FWK_NRT_014_004: gridded spectral integral parameter level-4 product based on level-3 Sentinel-1A and CFOSAT products.

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

b) General Information

Product Lines	WAVE_GLO_PHY_SPC-FWK_L4_NRT_014_004
Geographical coverage	Global: 180°W-180°E; 89°S – 90°N
Variables	The following integral parameters are used to describe observed swell fields along propagation time, from the storm source to the land, for a given month. The five most energetic swell fields are indicated. They are sorted independently for each lon/lat/time pixel. The letter "X" in the following fields stands for a number from 1 to 5, for first to fifth most energetic swell. **VAVH_SW[X]:** sea surface significant wave height (Hs), i.e. the mean wave height of the highest third of the waves (H1/3) for the considered swell system **VAVH_NBR_SW[X]:** number of measurements used for the estimation of the significant wave height (VAVH) **VPED_SW[X]:** sea surface wave from direction at partition spectral peak, i.e. wave principal direction at partition spectral peak / peak direction (Dp) **VPED_NBR_SW[X]:** number of measurements used for the estimation of the peak direction (VPED) **VTPK_SW[X]:** sea surface wave period at partition variance spectral density maximum, i.e. wave period at partition spectral peak / peak period (Tp) **VTPK_NBR_SW[X]:** number of measurements used for the estimation of the peak period (VTPK) **storm_number_SW[X]:** Corresponding index of the associated storm name in the storm_name variable. This identification name can be used to find the corresponding Level-3 product using the swell field format containing this ID in its filename **storm_name:** storm name composed of the storm date, longitude and latitude. This identification name can be used to find the corresponding Level-3 product using the swell field format containing this ID in its filename.
Product Type	Near Real Time – Level-4 observations
Available time series	Dec 2021- Present
Temporal resolution	3-hourly (one file contains 5 most energetic swell fields propagating from storm source until dissipation

	offshore or at the coast)
Update Frequency	Daily
Target delivery time	daily at 12 pm (noon) UTC
Horizontal resolution	1°x1°
Number of vertical levels	1
Format	NetCDF4

c) Production System Description

The SPeCtral FireWorKs system provides spectral integrated parameters of swell systems along their propagation path from the storm generation to the land. These parameters can be separated into observations and pseudo-observations. While the first ones are directly obtained from the wave spectrum partition of L2 satellite measurements, the pseudo-observations are derived using the 'Fireworks' algorithm. This algorithm propagates the observed partitions over the ocean using a linear wave propagation model from the observation time (both forwards and backwards). It then associates the propagated partitions by storm and filters out the partitions which do not pertain to any coherent swell system. The propagated partitions selected are then referred to as pseudo-observations.

WAVE_GLO_PHY_SPC-FWK_L4_NRT_014_004
Summary of the operational configuration: **Domain and resolution:** The gridded product merges L3 Sentinel-1A and CFOSAT swell field products over the global ocean. **Geographic coverage**: The geographic coverage is constrained by the availability of wave measurements propagating away from land and islands and in the absence of ice (updated daily from CMEMS SEAICE_GLO_SEAICE_L4_NRT_OBSERVATIONS_011_001 product). The measurements are provided over a regular spatial grid (1-degree resolution) with a regular time step (3h). The maps are updated and delivered daily. The VAVH, VPED and VTPK fields use the L3 swell measurements, accounting for their spatial proximity and data quality to provide wave information over each open ocean grid cell. See QUID for details.

d) Dataset details

Product Lines	WAVE_GLO_PHY_SPC-FWK_L4_NRT_014_004
Geographical coverage	Global
Datasets: cmems_obs-wave_glo_phy-spc-fwk_nrt_multi-l4-1deg_PT3H -i	
Product Type	Near Real Time – Level-4 observations
Update frequency	Daily
Available time series	Dec 2021 - Present
Target delivery time	Daily at 12pm (noon) UTC
Temporal resolution	3 hourly instantaneous

Dataset : cmems_obs-wave_glo_phy-spc-fwk_nrt_multi-l4-1deg_PT3H-i

Contains the integral wave parameters of the five most energetic swell fields, represented as gridded parameters.
VAVH_SW[X] [m] *(X stands for the partition number, from 0 to 4)* Partition significant wave height for swell X sea_surface_wave_significant_height
VAVH_NBR_SW[X] [-] Number of measurements used to estimate the partition significant wave height per lon/lat cell for swell X sea_surface_wave_significant_height_number_of_observations
VPED_SW[X] [deg] Partition peak direction (met. convention) for swell X sea_surface_wave_from_direction_at_variance_spectral_density_maximum
VPED_NBR_SW[X] [-] Number of measurements used to estimate the partition peak direction per lon/lat cell for swell X sea_surface_wave_from_direction_at_variance_spectral_density_maximum_number_of_observations
VTPK_SW[X] [s] Partition peak period for swell X sea_surface_wave_period_at_variance_spectral_density_maximum
VTPK_NBR_SW[X] [-] Number of measurements used to estimate the partition peak period per lon/lat cell for swell X sea_surface_wave_period_at_variance_spectral_density_maximum_number_of_observations
storm_number_SW[X] [-] Corresponding index of the storm in storm variable for swell X corresponding_storm_name_index
storm_name [-] storm_name

e) Additional information on some parameters

All Level-4 gridded products are delivered onto cartesian grids. All the variables are defined at the center of the output grid cell.

f) Processing information

At each delivery, the swell field analysis is performed over the last 3 weeks. This is to take into account the possible swell propagation from the storm source for more than 10 days before reaching the coast plus a potential storm duration of several days.

After this swell field analysis, the older swell fields are almost unchanged from one update to the next as very few observations can belong to the oldest swell fields. For the more recent days, new swell fields can appear from one day to the next if enough L2 observations have been gathered to create a new swell system. Also, older swell fields can be enriched with additional recent L2 observations.

All L4 products covering the swell field analysis time period (the last 3 weeks) are updated in this process.

g) Catalogue updating

The production chain runs once a day to produce level-3 SPC products from all latest available L2 measurements. The L4 production is then run from all available L3 products.
Level-4 files are updated by 12:00 UTC.

h) Time averaging

Instantaneous values are provided. There is no temporal averaging or interpolation performed.

i) Product use and impact

This product is related to the following Copernicus Marine Service product:

- WAVE_GLO_PHY_SPC-FWK_L3_NRT_014_002: Directional SPeCtral (SPC) integral parameter level-3 product based on observations of wave systems and their propagation over the ocean ('fireworks' algorithm).

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

All WAVE-TAC products (along-track level-3 and gridded level-4) are available through the Marine Data Store.

The nomenclature is described below for the original files.

cmems_obs-wave_glo_phy-spc_nrt_multi-l4-1deg_PT3H-<datemin>_<datemax>_P<dateprod>_<status>.nc

With:

- <datemin>= date of first measurement in the file YYYYmmddTHHMMZ
- <datemax>= date of last measurement in the file YYYYmmddTHHMMZ
- <dateprod>=date of production of the file YYYYmmddTHHMMZ
- <status> = "nrt" for the most recent files that will be subject to updates as more swell observations can be gathered to each swell fields. "rep" for the oldest versions.

4.1) FILE SIZE

DATASET NAME	NAME OF FILE	DIMENSION [GB]
cmems_obs-wave_glo_phy-spc-fwk_nrt_multi-l4-1deg_PT3H-i	cmems_obs-wave_glo_phy-spc_nrt_multi-l4-1deg_PT3H-<datemin>_<datemax>_P<dateprod>_<status>.nc	1.65

4.2) STRUCTURE OF FILES

Examples of the header of output NetCDF files are inserted in **annex**, for each dataset.

5) REFERENCES

Quality Information Document QUID: the associated QUID document for this product is CMEMS-WAVE-QUID-014-004.

6) ANNEX

Structure and semantic of NetCDF SPC level-4 files

Example of instantaneous NetCDF metadata from these files is given below.

```
netcdf cmems_obs-wave_glo_phy-spc_nrt_multi-l4-1deg_PT3H-20210101T0000Z_20210131T0000Z_P20211123T1324Z-nrt {
dimensions:
        longitude = 360 ;
        latitude = 132 ;
        time = 195 ;
        storm_number = 40 ;
        name_size = 39 ;
variables:
        int VAVH_SW0(time, latitude, longitude) ;
                VAVH_SW0:_FillValue = -999 ;
                VAVH_SW0:units = "m" ;
                VAVH_SW0:long_name = "Partition significant wave height for swell 0" ;
                VAVH_SW0:standard_name = "sea_surface_wave_significant_height" ;
                VAVH_SW0:description = "VAVH estimated by interpolating the L3 data on a regular lon/lat grid for swell 0. The gridded partition significant wave height is estimated from Level-3 spectral products, accounting for their spatial, temporal proximity and data quality." ;
                VAVH_SW0:scale_factor = 0.001 ;
        int VAVH_NBR_SW0(time, latitude, longitude) ;
                VAVH_NBR_SW0:units = "1" ;
                VAVH_NBR_SW0:long_name = "Number of measurements used to estimate the partition significant wave height per lon/lat cell for swell 0" ;
                VAVH_NBR_SW0:standard_name = "sea_surface_wave_significant_height_number_of_observations" ;
                VAVH_NBR_SW0:description = "Number of measurements used to estimate the gridded partition significant wave height in the lon/lat cell for swell 0" ;
        int VTPK_SW0(time, latitude, longitude) ;
                VTPK_SW0:_FillValue = -999 ;
                VTPK_SW0:units = "s" ;
                VTPK_SW0:long_name = "Partition peak period for swell 0" ;
                VTPK_SW0:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum" ;
                VTPK_SW0:description = "VTPK estimated by interpolating the L3 data on a regular lon/lat grid for swell 0. The gridded partition peak period is estimated from Level-3 spectral products, accounting for their spatial, temporal proximity and data quality." ;
                VTPK_SW0:scale_factor = 0.01 ;
        int VTPK_NBR_SW0(time, latitude, longitude) ;
                VTPK_NBR_SW0:units = "1" ;
                VTPK_NBR_SW0:long_name = "Number of measurements used to estimate the partition peak period per lon/lat cell for swell 0" ;
                VTPK_NBR_SW0:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum_number_of_observations" ;
                VTPK_NBR_SW0:description = "Number of measurements used to estimate the gridded partition peak period in the lon/lat cell for swell 0" ;
        int VPED_SW0(time, latitude, longitude) ;
                VPED_SW0:_FillValue = -999 ;
                VPED_SW0:units = "degree" ;
                VPED_SW0:long_name = "Partition peak direction (met. convention) for swell 0" ;
                VPED_SW0:standard_name = "sea_surface_wave_from_direction_at_variance_spectral_density_maximum" ;
                VPED_SW0:description = "VPED estimated by interpolating the L3 data on a regular lon/lat grid for swell 0. The gridded partition peak direction is estimated from Level-3 spectral products, accounting for their spatial, temporal proximity and data quality." ;
                VPED_SW0:scale_factor = 0.1 ;
        int VPED_NBR_SW0(time, latitude, longitude) ;
                VPED_NBR_SW0:units = "1" ;
```

VPED_NBR_SW0:long_name = "Number of measurements used to estimate the partition peak direction per lon/lat cell for swell 0" ;
VPED_NBR_SW0:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum_number_of_observations" ;
VPED_NBR_SW0:description = "Number of measurements used to estimate the gridded partition peak direction in the lon/lat cell for swell 0" ;
int storm_number_SW0(time, latitude, longitude) ;
storm_number_SW0:units = "1" ;
storm_number_SW0:long_name = "Corresponding index of the storm in storm variable for swell 0" ;
storm_number_SW0:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum_number_of_observations" ;
storm_number_SW0:description = "Corresponding index of the associated storm in storm variable for swell 0" ;
int VAVH_SW1(time, latitude, longitude) ;
VAVH_SW1:_FillValue = -999 ;
VAVH_SW1:units = "m" ;
VAVH_SW1:long_name = "Partition significant wave height for swell 1" ;
VAVH_SW1:standard_name = "sea_surface_wave_significant_height" ;
VAVH_SW1:description = "VAVH estimated by interpolating the L3 data on a regular lon/lat grid for swell 1. The gridded partition significant wave height is estimated from Level-3 spectral products, accounting for their spatial, temporal proximity and data quality." ;
VAVH_SW1:scale_factor = 0.001 ;
int VAVH_NBR_SW1(time, latitude, longitude) ;
VAVH_NBR_SW1:units = "1" ;
VAVH_NBR_SW1:long_name = "Number of measurements used to estimate the partition significant wave height per lon/lat cell for swell 1" ;
VAVH_NBR_SW1:standard_name = "sea_surface_wave_significant_height_number_of_observations" ;
VAVH_NBR_SW1:description = "Number of measurements used to estimate the gridded partition significant wave height in the lon/lat cell for swell 1" ;
int VTPK_SW1(time, latitude, longitude) ;
VTPK_SW1:_FillValue = -999 ;
VTPK_SW1:units = "s" ;
VTPK_SW1:long_name = "Partition peak period for swell 1" ;
VTPK_SW1:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum" ;
VTPK_SW1:description = "VTPK estimated by interpolating the L3 data on a regular lon/lat grid for swell 1. The gridded partition peak period is estimated from Level-3 spectral products, accounting for their spatial, temporal proximity and data quality." ;
VTPK_SW1:scale_factor = 0.01 ;
int VTPK_NBR_SW1(time, latitude, longitude) ;
VTPK_NBR_SW1:units = "1" ;
VTPK_NBR_SW1:long_name = "Number of measurements used to estimate the partition peak period per lon/lat cell for swell 1" ;
VTPK_NBR_SW1:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum_number_of_observations" ;
VTPK_NBR_SW1:description = "Number of measurements used to estimate the gridded partition peak period in the lon/lat cell for swell 1" ;
int VPED_SW1(time, latitude, longitude) ;
VPED_SW1:_FillValue = -999 ;
VPED_SW1:units = "degree" ;
VPED_SW1:long_name = "Partition peak direction (met. convention) for swell 1" ;
VPED_SW1:standard_name = "sea_surface_wave_from_direction_at_variance_spectral_density_maximum" ;
VPED_SW1:description = "VPED estimated by interpolating the L3 data on a regular lon/lat grid for swell 1. The gridded partition peak direction is estimated from Level-3 spectral products, accounting for their spatial, temporal proximity and data quality." ;
VPED_SW1:scale_factor = 0.1 ;
int VPED_NBR_SW1(time, latitude, longitude) ;
VPED_NBR_SW1:units = "1" ;
VPED_NBR_SW1:long_name = "Number of measurements used to estimate the partition peak direction per lon/lat cell for swell 1" ;
VPED_NBR_SW1:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum_number_of_observations" ;

VPED_NBR_SW1:description = "Number of measurements used to estimate the gridded partition peak direction in the lon/lat cell for swell 1" ;
int storm_number_SW1(time, latitude, longitude) ;
storm_number_SW1:units = "1" ;
storm_number_SW1:long_name = "Corresponding index of the storm in storm variable for swell 1" ;
storm_number_SW1:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum_number_of_observations" ;
storm_number_SW1:description = "Corresponding index of the associated storm in storm variable for swell 1" ;
int VAVH_SW2(time, latitude, longitude) ;
VAVH_SW2:_FillValue = -999 ;
VAVH_SW2:units = "m" ;
VAVH_SW2:long_name = "Partition significant wave height for swell 2" ;
VAVH_SW2:standard_name = "sea_surface_wave_significant_height" ;
VAVH_SW2:description = "VAVH estimated by interpolating the L3 data on a regular lon/lat grid for swell 2. The gridded partition significant wave height is estimated from Level-3 spectral products, accounting for their spatial, temporal proximity and data quality." ;
VAVH_SW2:scale_factor = 0.001 ;
int VAVH_NBR_SW2(time, latitude, longitude) ;
VAVH_NBR_SW2:units = "1" ;
VAVH_NBR_SW2:long_name = "Number of measurements used to estimate the partition significant wave height per lon/lat cell for swell 2" ;
VAVH_NBR_SW2:standard_name = "sea_surface_wave_significant_height_number_of_observations" ;
VAVH_NBR_SW2:description = "Number of measurements used to estimate the gridded partition significant wave height in the lon/lat cell for swell 2" ;
int VTPK_SW2(time, latitude, longitude) ;
VTPK_SW2:_FillValue = -999 ;
VTPK_SW2:units = "s" ;
VTPK_SW2:long_name = "Partition peak period for swell 2" ;
VTPK_SW2:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum" ;
VTPK_SW2:description = "VTPK estimated by interpolating the L3 data on a regular lon/lat grid for swell 2. The gridded partition peak period is estimated from Level-3 spectral products, accounting for their spatial, temporal proximity and data quality." ;
VTPK_SW2:scale_factor = 0.01 ;
int VTPK_NBR_SW2(time, latitude, longitude) ;
VTPK_NBR_SW2:units = "1" ;
VTPK_NBR_SW2:long_name = "Number of measurements used to estimate the partition peak period per lon/lat cell for swell 2" ;
VTPK_NBR_SW2:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum_number_of_observations" ;
VTPK_NBR_SW2:description = "Number of measurements used to estimate the gridded partition peak period in the lon/lat cell for swell 2" ;
int VPED_SW2(time, latitude, longitude) ;
VPED_SW2:_FillValue = -999 ;
VPED_SW2:units = "degree" ;
VPED_SW2:long_name = "Partition peak direction (met. convention) for swell 2" ;
VPED_SW2:standard_name = "sea_surface_wave_from_direction_at_variance_spectral_density_maximum" ;
VPED_SW2:description = "VPED estimated by interpolating the L3 data on a regular lon/lat grid for swell 2. The gridded partition peak direction is estimated from Level-3 spectral products, accounting for their spatial, temporal proximity and data quality." ;
VPED_SW2:scale_factor = 0.1 ;
int VPED_NBR_SW2(time, latitude, longitude) ;
VPED_NBR_SW2:units = "1" ;
VPED_NBR_SW2:long_name = "Number of measurements used to estimate the partition peak direction per lon/lat cell for swell 2" ;
VPED_NBR_SW2:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum_number_of_observations" ;
VPED_NBR_SW2:description = "Number of measurements used to estimate the gridded partition peak direction in the lon/lat cell for swell 2" ;
int storm_number_SW2(time, latitude, longitude) ;
storm_number_SW2:units = "1" ;

storm_number_SW2:long_name = "Corresponding index of the storm in storm variable for swell 2" ;
storm_number_SW2:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum_number_of_observations" ;
storm_number_SW2:description = "Corresponding index of the associated storm in storm variable for swell 2" ;
int VAVH_SW3(time, latitude, longitude) ;
VAVH_SW3:_FillValue = -999 ;
VAVH_SW3:units = "m" ;
VAVH_SW3:long_name = "Partition significant wave height for swell 3" ;
VAVH_SW3:standard_name = "sea_surface_wave_significant_height" ;
VAVH_SW3:description = "VAVH estimated by interpolating the L3 data on a regular lon/lat grid for swell 3. The gridded partition significant wave height is estimated from Level-3 spectral products, accounting for their spatial, temporal proximity and data quality." ;
VAVH_SW3:scale_factor = 0.001 ;
int VAVH_NBR_SW3(time, latitude, longitude) ;
VAVH_NBR_SW3:units = "1" ;
VAVH_NBR_SW3:long_name = "Number of measurements used to estimate the partition significant wave height per lon/lat cell for swell 3" ;
VAVH_NBR_SW3:standard_name = "sea_surface_wave_significant_height_number_of_observations" ;
VAVH_NBR_SW3:description = "Number of measurements used to estimate the gridded partition significant wave height in the lon/lat cell for swell 3" ;
int VTPK_SW3(time, latitude, longitude) ;
VTPK_SW3:_FillValue = -999 ;
VTPK_SW3:units = "s" ;
VTPK_SW3:long_name = "Partition peak period for swell 3" ;
VTPK_SW3:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum" ;
VTPK_SW3:description = "VTPK estimated by interpolating the L3 data on a regular lon/lat grid for swell 3. The gridded partition peak period is estimated from Level-3 spectral products, accounting for their spatial, temporal proximity and data quality." ;
VTPK_SW3:scale_factor = 0.01 ;
int VTPK_NBR_SW3(time, latitude, longitude) ;
VTPK_NBR_SW3:units = "1" ;
VTPK_NBR_SW3:long_name = "Number of measurements used to estimate the partition peak period per lon/lat cell for swell 3" ;
VTPK_NBR_SW3:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum_number_of_observations" ;
VTPK_NBR_SW3:description = "Number of measurements used to estimate the gridded partition peak period in the lon/lat cell for swell 3" ;
int VPED_SW3(time, latitude, longitude) ;
VPED_SW3:_FillValue = -999 ;
VPED_SW3:units = "degree" ;
VPED_SW3:long_name = "Partition peak direction (met. convention) for swell 3" ;
VPED_SW3:standard_name = "sea_surface_wave_from_direction_at_variance_spectral_density_maximum" ;
VPED_SW3:description = "VPED estimated by interpolating the L3 data on a regular lon/lat grid for swell 3. The gridded partition peak direction is estimated from Level-3 spectral products, accounting for their spatial, temporal proximity and data quality." ;
VPED_SW3:scale_factor = 0.1 ;
int VPED_NBR_SW3(time, latitude, longitude) ;
VPED_NBR_SW3:units = "1" ;
VPED_NBR_SW3:long_name = "Number of measurements used to estimate the partition peak direction per lon/lat cell for swell 3" ;
VPED_NBR_SW3:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum_number_of_observations" ;
VPED_NBR_SW3:description = "Number of measurements used to estimate the gridded partition peak direction in the lon/lat cell for swell 3" ;
int storm_number_SW3(time, latitude, longitude) ;
storm_number_SW3:units = "1" ;
storm_number_SW3:long_name = "Corresponding index of the storm in storm variable for swell 3" ;
storm_number_SW3:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum_number_of_observations" ;

storm_number_SW3:description = "Corresponding index of the associated storm in storm variable for swell 3" ;
int VAVH_SW4(time, latitude, longitude) ;
VAVH_SW4:_FillValue = -999 ;
VAVH_SW4:units = "m" ;
VAVH_SW4:long_name = "Partition significant wave height for swell 4" ;
VAVH_SW4:standard_name = "sea_surface_wave_significant_height" ;
VAVH_SW4:description = "VAVH estimated by interpolating the L3 data on a regular lon/lat grid for swell 4. The gridded partition significant wave height is estimated from Level-3 spectral products, accounting for their spatial, temporal proximity and data quality." ;
VAVH_SW4:scale_factor = 0.001 ;
int VAVH_NBR_SW4(time, latitude, longitude) ;
VAVH_NBR_SW4:units = "1" ;
VAVH_NBR_SW4:long_name = "Number of measurements used to estimate the partition significant wave height per lon/lat cell for swell 4" ;
VAVH_NBR_SW4:standard_name = "sea_surface_wave_significant_height_number_of_observations" ;
VAVH_NBR_SW4:description = "Number of measurements used to estimate the gridded partition significant wave height in the lon/lat cell for swell 4" ;
int VTPK_SW4(time, latitude, longitude) ;
VTPK_SW4:_FillValue = -999 ;
VTPK_SW4:units = "s" ;
VTPK_SW4:long_name = "Partition peak period for swell 4" ;
VTPK_SW4:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum" ;
VTPK_SW4:description = "VTPK estimated by interpolating the L3 data on a regular lon/lat grid for swell 4. The gridded partition peak period is estimated from Level-3 spectral products, accounting for their spatial, temporal proximity and data quality." ;
VTPK_SW4:scale_factor = 0.01 ;
int VTPK_NBR_SW4(time, latitude, longitude) ;
VTPK_NBR_SW4:units = "1" ;
VTPK_NBR_SW4:long_name = "Number of measurements used to estimate the partition peak period per lon/lat cell for swell 4" ;
VTPK_NBR_SW4:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum_number_of_observations" ;
VTPK_NBR_SW4:description = "Number of measurements used to estimate the gridded partition peak period in the lon/lat cell for swell 4" ;
int VPED_SW4(time, latitude, longitude) ;
VPED_SW4:_FillValue = -999 ;
VPED_SW4:units = "degree" ;
VPED_SW4:long_name = "Partition peak direction (met. convention) for swell 4" ;
VPED_SW4:standard_name = "sea_surface_wave_from_direction_at_variance_spectral_density_maximum" ;
VPED_SW4:description = "VPED estimated by interpolating the L3 data on a regular lon/lat grid for swell 4. The gridded partition peak direction is estimated from Level-3 spectral products, accounting for their spatial, temporal proximity and data quality." ;
VPED_SW4:scale_factor = 0.1 ;
int VPED_NBR_SW4(time, latitude, longitude) ;
VPED_NBR_SW4:units = "1" ;
VPED_NBR_SW4:long_name = "Number of measurements used to estimate the partition peak direction per lon/lat cell for swell 4" ;
VPED_NBR_SW4:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum_number_of_observations" ;
VPED_NBR_SW4:description = "Number of measurements used to estimate the gridded partition peak direction in the lon/lat cell for swell 4" ;
int storm_number_SW4(time, latitude, longitude) ;
storm_number_SW4:units = "1" ;
storm_number_SW4:long_name = "Corresponding index of the storm in storm variable for swell 4" ;
storm_number_SW4:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum_number_of_observations" ;
storm_number_SW4:description = "Corresponding index of the associated storm in storm variable for swell 4" ;
double time(time) ;
time:axis = "T" ;

```
                time:units = "days since 1950-01-01T00:00:00Z" ;
                time:long_name = "time" ;
                time:standard_name = "time" ;
        float longitude(longitude) ;
                longitude:axis = "X" ;
                longitude:units = "degrees_east" ;
                longitude:long_name = "longitude" ;
                longitude:standard_name = "longitude" ;
                longitude:valid_min = -180.f ;
                longitude:valid_max = 180.f ;
        float latitude(latitude) ;
                latitude:axis = "Y" ;
                latitude:units = "degrees_north" ;
                latitude:long_name = "latitude" ;
                latitude:standard_name = "latitude" ;
                latitude:valid_min = -89.f ;
                latitude:valid_max = 89.f ;
        char storm_name(storm_number, name_size) ;
                storm_name:long_name = "storm name" ;
                storm_name:standard_name = "storm name" ;

// global attributes:
        :Software_version = "L4NRT_SPC:1.5.14.dev0" ;
        :Comment = "Spectral wave parameters along propagation projected on a regular space-time grid for multiple swell
        field and using Level-3 spectral products" ;
        :Conventions = "CF-1.6" ;
        :Metadata_Conventions = "Unidata Dataset Discovery v1.0" ;
        :cdm_data_type = "Grid" ;
        :creator_email = "servicedesk.cmems@mercator-ocean.eu" ;
        :references = "https://marine.copernicus.eu/fr/a-propos/producteurs/wave-tac " ;
        :creator_url = "http://marine.copernicus.eu" ;
        :project = "COPERNICUS MARINE ENVIRONMENT MONITORING SERVICE (CMEMS)" ;
        :contact = "servicedesk.cmems@mercator-ocean.eu" ;
        :processing_level = "L4" ;
        :creation_date = "20211123T1324Z" ;
        :keywords_vocabulary = "NetCDF COARDS Climate and Forecast Standard Names" ;
        :institution = "CLS" ;
        :creator_name = "CMEMS - Wave Thematic Assembly Center" ;
        :license = "http://marine.copernicus.eu/web/27-service-commitments-and-licence.php" ;
        :standard_name_vocabulary = "NetCDF Climate and Forecast (CF) Metadata Convention Standard Name Table v37"
        ;
        :title = "NRT merged Global Ocean Gridded spectral parameters L4 product and derived variables" ;
        :source_platforms = "[\'Sentinel-1A\', \'CFOSAT\'] measurements" ;
        :product_version = "202112" ;
        :based_on = "[\'Sentinel-1A\', \'CFOSAT\'] WV L3 SPC products" ;
        :time_coverage_start = "20210101T0000Z" ;
        :time_coverage_end = "20210131T2359Z" ;
        :time_coverage_resolution = "3H" ;
}
```